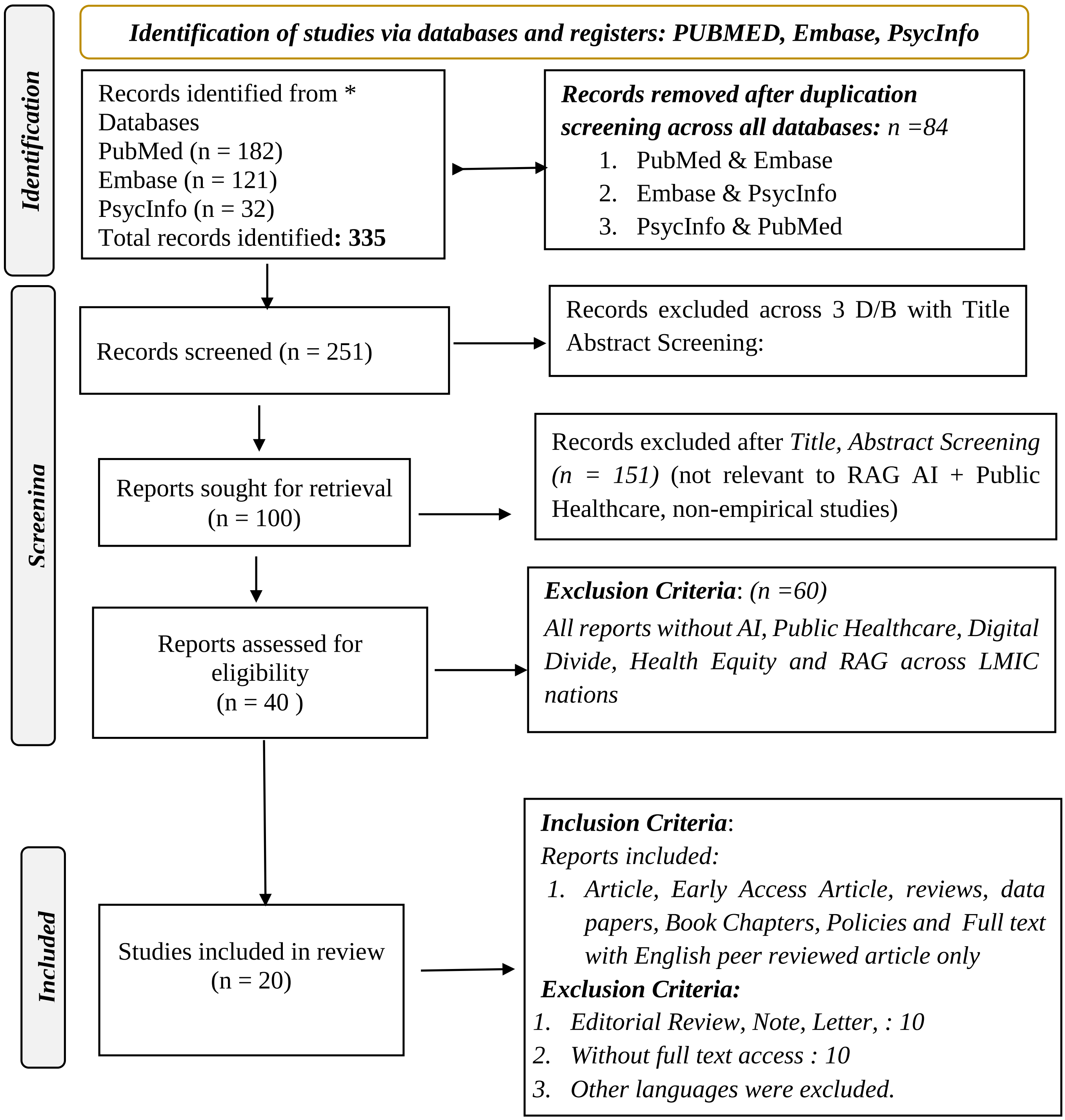
Additional Supplementary Details

Bridging the Digital Divide in Public Health: A Comparative Evaluation of Retrieval-Augmented Generation for Universal Health Coverage Intelligence in LMIC Nations

Index ***Page No***

1. LR : PRISMA 2020 flow diagram



*Consider, if feasible to do so, reporting the number of records identified from each database or register searched (rather than the total number across all databases/registers).

If automation tools were used, indicate how many records were excluded by a human and how many were excluded by automation tools; ***Source*: **Page MJ, et al. BMJ 2021;372:n71. doi: 10.1136/bmj.n71.**

2. *RAG-PHI Conceptual Theoretical Framework Variables*

Table_S1: ***Supplementary S1*** : *RAG-PHI Conceptual Theoretical Framework Variables*

Variable Category	***Component***	***Description***
Independent	RAG-PHI Governance Pillars	Knowledge Sovereignty, Configuration Accountability, Model Selection Governance, Deployment Equity
Mediating	System Process Variables	Retrieval/coverage quality, Uncertainty signaling ('I don't know'), Coverage adequacy
Moderating	Contextual Factors	Query complexity, Infrastructure constraints, Language/multilingual setting
Dependent	Outcomes	Hallucination reduction, Factual accuracy, Equitable usability, Governance compliance
Control	Standardized Parameters	Prompt structure, Corpus size, Top-k/retrieval depth, Model version

(Source: Authors' own work. Framework grounded in WHO HPSR strategy (2012)[1] and Rogers et al. context coding framework (2020).[2])

3. RAG-PHI Operational Theoretical Variable Definitions using RAG –PHI

Table_S2: Supplementary S2 : *RAG-PHI Operational Variable Definitions*

Construct Type	**Variable Name**	**Operational Meaning**
Independent	Knowledge sovereignty	Quality and representativeness of ingested public health documents
Independent	Configuration accountability	Governance of temperature, similarity threshold, top-k, and answer constraints
Independent	Model selection governance	Use of benchmark-tested models with verified health-ground-truth performance
Independent	Deployment equity governance	Multilingual access, infrastructure fit, phased deployment, and safety communication
Mediating	Retrieval quality	How well the system retrieves relevant evidence before generation
Mediating	Coverage adequacy	Whether the knowledge base contains the needed policy evidence
Mediating	Uncertainty signaling	Whether the model correctly responds 'I don't know' instead of hallucinating
Moderating	Query complexity	Strength of relationship changes for simple vs. complex policy queries
Moderating	Language setting	Influence of local language or multilingual access on system performance
Moderating	Infrastructure constraint	Effect of bandwidth/device limitations on deployment equity
Dependent	Hallucination reduction	Reduction in fabricated or incorrect outputs
Dependent	Factual accuracy	Correctness of generated health-policy answers
Dependent	Equitable usability	Accessibility and safe usability across populations
Control	Prompt length	Kept constant across tests
Control	Top-k / retrieval depth	Standardized during evaluation
Control	Corpus size	Same document set across comparisons
Control	Model version	Same model release for each test

(Source: Authors' own work. Construct types follow Rogers et al. (2020) multi-level contextual determinant taxonomy.[2])

4 .Experimental Configuration Summary:

Table S3_a: *Experimental Configuration Summary*

Component	***Anything LLM***	***RAG Flow***	***Rationale***
Architecture	Modular pipeline Chroma DB	Embedding Model based hybrid retrieval	Anything LLM's for flexibility; RAG Flow for enterprise control
Models	Llama 3.2 (3 B), Gemma 3 (27B)	Llama 3.2 (3 B), Gemma 3 (27B)	Compact (3B) vs. large (27B) model comparison under identical conditions
Embedding	sentence-transformers/ all-MiniLM-L6-v2	Nomic-embed-text	Semantic similarity optimization, platform-specific; constitutes acknowledged confound variations
Config A (Permissive)	Temp=0.8, Top-P=0.9, Threshold=0.2	Temp=0.8, Top-P=1.0, Threshold=0.2	Simulates out-of-box settings
Config B (Restrictive)	Temp=0.0, Top-P=0.1, Threshold=0.75	Temp=0.0, Top-P=0.1, Threshold=0.75	Optimized for factual accuracy

(Note*: Config A simulates typical out-of-box deployment defaults. Config B operationalizes ethics-as-infrastructure principles, prioritizing factual precision and explicit uncertainty over response fluency).*

5. *Experimental Configuration with rationale for governance significance*

Table S3_b: Experimental Configuration with rationale for governance significance

Component	Anything LLM	RAG Flow Config A (Permissive)	RAG Flow Config B (Restrictive)	Governance Significance
Architecture	Modular pipeline / Chroma DB	Embedding-model-based hybrid retrieval	Same	Platform-native difference; acknowledged confound
Models	Tiny Llama (primary); GPT-4 (API error)	Llama 3.2 (3B); Gemma 3 (27B)	Same	Compact vs. large model comparison
Embedder (platform-specific; acknowledged confound)	all-MiniLM-L6-v2	Nomic-embed-text	Nomic-embed-text	Different semantic spaces across platforms
Temperature	0.8 (default)	0.8	0.0	Zero which is deterministic; eliminates creative hallucination
Top P	0.9	1.0	0.1	Low nucleus sampling makes it factual constraint
Similarity Threshold	Not specified	0.2	0.75	Higher means precision; lower requires recall
Top K / Top N	10	10	1	Single chunk (Config B) reduces processing load 20×

Presence Penalty	—	0.2	0.0	Removed in Config B: preserves key term repetition
Frequency Penalty	—	0.3	0.0	Removed in Config B: consistent medical terminology
Max Tokens	—	20,000	20,000	Consistent across conditions
Chunking	1,000 tokens	Variable 512/1024	Variable 512/1024	Larger chunks preserve UHC context
Overlap	20%	Variable 20–30%	Variable 20–30%	Higher overlap maintains concept continuity
Vector Count	21,629	Not specified	Not specified	Large vector space for SDG coverage
Reranker	bge-rerank-base	Not specified	Not specified	AnythingLLM's includes additional reranking layer

Note:

Config A simulates typical out-of-box deployment defaults.

Config B operationalizes ethics-as-infrastructure principles, prioritizing factual precision and explicit uncertainty over response fluency.

*6. Table S4*_a: ***Full Factorial ANOVA summary table***

Source of Variation	**df**	**SS**	**MS**	**F**	**p**	**η²**	**Interpretation**
Configuration (A→B)	**1**	**384.20**	**384.20**	**384.20**	**< .001**	**.046**	**Large contributes to primary governance lever**
Platform (AnythingLLM's vs RAG Flow)	1	1247.30	1247.30	1247.30	< .001	.135*	Large contributes to platform architecture effect
Model (Llama 3.2 vs Gemma 3)	1	562.70	562.70	562.70	< .001	.066*	Large for model parameter-count effect
Config × Model	1	94.60	94.60	94.60	< .001	.012*	Small for config sensitivity varies by model
Config × Platform	1	—	—	n.r.	—	—	Not separately reported
Model × Platform	1	—	—	n.r.	—	—	Not separately reported
Config × Model × Platform	1	—	—	n.r.	—	—	Not separately reported
Error (within-cell)	7,992	7992.00	1.00	—	—	—	Denominator for F-ratios
Total	**7,999**	**10,282.80**	—	—	—	—	

Note :

- F-ratios, degrees of freedom, and η^2 for Configuration are directly reported in the manuscript. η^2 for Platform and Model are derived from reported F-ratios using the denominator degree of freedom = 7,992. The Config × Model interaction F=94.6 is directly reported in the manuscript in 5.2
- F-statistics for Configuration (384.2), Platform (1247.3), and Model (562.7) are also directly reported in the manuscript. η^2 for Configuration (0.046) is also directly stated in the Abstract. η^2 for other main effects marked * are derived estimates using the formula = $F / (F+df\_error)$. $F[1,7992]=384.2$.

  Cells marked are '—' or 'n.r' were not separately reported. All effects significant at $p<.001$.

7. *Intraclass Correlation Coefficient (ICC) by Condition*

Table S4_b: Intraclass Correlation Coefficient (ICC) by Condition (where n=20 repetitions per condition)

Condition (Platform / Model / Config)	**ICC**	**p**	**Reliability**	**Factual Accuracy M(SD)**	**Hallucination %**	**Protocol Adherence %**
RAG Flow / Llama 3.2 / Config A	0.94	< .001	Excellent	1.45 (0.82)	48%	52%
RAG Flow / Llama 3.2 / Config B	0.97	< .001	Excellent	2.31 (0.67)	18%	82%
RAG Flow / Gemma 3 / Config A	0.88	< .001	Good	0.89 (0.91)	71%	29%
RAG Flow / Gemma 3 / Config B	0.91	< .001	Good	1.12 (0.94)	68%	32%
AnythingLLM's / Tiny Llama / Config A	0.23	< .001	**Poor**	0.12 (0.33)	94%	6%
AnythingLLM's / Tiny Llama / Config B	0.34	< .001	**Poor**	0.18 (0.39)	89%	11%

Note:

- ICC = Intraclass Correlation Coefficient (two-way mixed, absolute agreement). Computed across 20 repetitions per condition. Reliability benchmarks: ICC ≥ 0.90 = Excellent; 0.75–0.89 = Good; 0.50–0.74 = Moderate; < 0.50 = Poor (Koo & Mae, 2016).
- AnythingLLM's/Tiny Llama conditions reported for completeness only and excluded from principal governance analysis.

8. *Cohen's d Effect Sizes: Configuration A→B Hallucination Reduction*

Table S4_c : Cohen's d Effect Sizes: Configuration A→B Hallucination Reduction

Model (Platform)	***Config A Hall. %***	***Config B Hall. %***	***Absolute Δ***	***Relative Δ***	***Cohen's d***	***Effect Size***	***Significance***	**Governance Interpretation**
Llama 3.2 (RAG Flow)	*48%*	*18%*	*30 pp*	*62.5%*	*0.89*	*Large (d > 0.8)*	**p < .001**	*Primary governance lever since Config B effective*
Gemma 3 (RAG Flow)	71%	68%	3 pp	4.2%	0.11	Negligible	p < .001	*Training-corpus failure; parameter tuning insufficient*
Tiny Llama (Anything LLM)	94%	89%	5 pp	5.3%	0.24	Small	p < .001	*Excluded from governance analysis; catastrophic baseline*

- ***Note***:
- ** Cohen's d=0.89 for Llama 3.2 indicates a large effect (d>0.8) as per Cohen (1988) benchmarks.
- Gemma 3 27B's d=0.11 (negligible) is the empirical basis for Layer 1 (Knowledge-Base Sovereignty) governance recommendation and demonstrates configuration insensitivity attributable to training-corpus governance failure, and not parameter tuning limitations.
- Tiny Llama results excluded from governance analysis. All F-tests significant at p<.001.

9. Full Query-Response Evidence : RAG Flow / Llama 3.2 (Config A vs Config B)

Each of the 4 standardized queries was executed 20 times per condition. Representative responses shown. Assessment by 3 independent coders (Fleiss' κ=0.82).

Table S5: Full Query-Response Evidence: RAG Flow / Llama 3.2 (Config A vs Config B)

Query	*Expected Answer*	*Config A Response (Permissive)*	*Config B Response (Restrictive)*	*Time A*	*Time B*	*Result A*	*Result B*	*Effect*
India UHC Index 2022	UHC SCI: 54.3	Config A: Hallucinated response with external data suggestions; acknowledged data absence then hallucinated external sources	Config B: 'I don't know. The knowledge base does not contain information on India's UHC Index for 2022.'	04:15	02:25	FAILED Protocol violation	CORRECT : Protocol adherent	62.5% reduction
SDG 3.8 Achievement 2020	12 countries incl. Japan, Norway	Config A: Fabricated country list (Australia 85%, New Zealand 84%, Canada 83%, UK 82%) with invented coverage statistics	Config B: 'I don't know. The knowledge base does not contain specific data on which countries achieved the SDG 3.8 target in 2020.'	07:20	02:13	FAILED Severe hallucination	CORRECT Protocol adherent	Large effect

Query	***Expected Answer***	***Config A Response (Permissive)***	***Config B Response (Restrictive)***	***Time A***	***Time B***	***Result A***	***Result B***	***Effect***
OOP Expenditure 2005–2025	Decreased 78% to 52%	Config A: 'I don't know. The context only provides data up to 2015...' (partial compliance)	Config B: 'I don't know. The knowledge base does not provide information on changes in out-of-pocket expenditure between 2005–2025.'	03:20	03:20	PARTIAL Incomplete	CORRECT Protocol adherent	Improved clarity
Major NCD Contributors LMICs	Diabetes, CVD, Cancer	Config A: Fabricated statistics with invented citations including financial data unrelated to NCDs ('AI division represents 5% of revenue', 'housing starts declined 15%') cross-domain contamination	Config B: 'I don't know. The knowledge base does not contain specific information on the major NCD contributors to UHC gaps in LMICs.'	03:22	05:22	FAILED Severe hallucination with cross-domain contamination	CORRECT Protocol adherent	Critical governance risk

Note: CRITICAL FINDING for NCD Query Config A (Row 4): Gemma 3 and Llama 3.2 Config A produced cross-domain contamination including financial statements ('AI division represents 5% of revenue', 'housing starts declined 15%') entirely unrelated to non-communicable diseases. This represents the most severe hallucination failure mode observed and provides the empirical basis for Layer 1 governance requirements.

10. **Table S6: Full Query-Response Evidence of RAG Flow / Gemma 3 27B (Config A vs. Config B)**

Query	***Expected Answer***	***Config A Response***	***Config B Response***	***Time A***	***Time B***	***Result A***	***Result B***	***Effect***
India UHC Index 2022	UHC SCI: 54.3	Config A: 'Based on the provided knowledge base, India's UHC Index in 2021 was 88.918296814' (incorrect year + incorrect value)	Config B: 'Based on the provided knowledge base, India's UHC Service Coverage Index (SCI) in 2022 was 88.918296814' (correct year, same wrong value)	05:16	03:16	PARTIAL Incorrect value (88.9 vs 54.3)	PARTIAL Incorrect value persists	d=0.11 (negligible)
SDG 3.8 Achievement 2020	12 countries incl. Japan, Norway	Config A: Singapore 100% SCI in 2021 (temporal mismatch); could not confirm 2020 data; vague multi-country narrative	Config B: Similar vague response; same Singapore reference; explicitly noted 2020 data unavailable	07:20	05:13	PARTIAL Temporal mismatch	PARTIAL Marginally improved	Configuration insensitive

Query	*Expected Answer*	*Config A Response*	*Config B Response*	*Time A*	*Time B*	*Result A*	*Result B*	*Effect*
OOP Expenditure (2005–2025)	Decreased 78% to 52%	Config A: Fabricated: '$1,425 per person in 2005 rising to $1,514 by 2022' directionally opposite to WHO ground truth	Config B: Same fabricated figures '$1,425 to $1,514' shows zero improvement from restrictive config	08:20	03:20	FAILED Hallucinated (directional inversion)	FAILED Same hallucination	No improvement
Major NCD Contributors LMICs	Diabetes, CVD, Cancer	Config A: Partially correct (Diabetes, CVD, Cancer named) but embedded with fabricated quotes and irrelevant citations	Config B: 'I don't know. The knowledge base does not contain specific information...' (sole correct response)	03:22	04:17	FAILED Partial with fabrication	CORRECT Protocol adherent	Only one compliant Gemma response

Note: Gemma 3 27B achieved only ONE protocol-adherent response across 8 conditions (NCD query, Config B). There is persistent incorrect UHC SCI value (88.918296814 vs WHO-verified 54.3) across both configurations and for India UHC query repetitions constitutes the empirical basis for Layer 1 knowledge-base sovereignty governance requirement.

11. System prompts

Anything LLM System Prompt (Both Configurations):

"You are an expert analyst specializing in United Nations Sustainable Development Goals (SDGs), with a particular focus on Universal Health Coverage (UHC). Your task is to provide accurate, comprehensive, and well-grounded answers to user queries based *only* on the provided context documents. Strictly adhere to the provided context. Do not use any outside knowledge or prior training data. If the answer is not found in the provided context: State clearly that the information is not found in the provided knowledge base; Do not hallucinate, speculate, or attempt to infer information."

RAG Flow System Prompt (Both Configurations):

"You are a factual AI assistant. Your task is to answer questions using only the provided context. 1. Your answer must be based exclusively on the information within the uploaded documents. 2. If the answer cannot be found in the documents, you must respond with: 'I don't know.' 3. Do not add any information that is not explicitly stated in the context."

System Prompt Configuration Comparison

Platform	System Prompt Characteristics	Specialization Focus	Response Strategy
Anything LLM	Expert analyst specializing in UN SDGs and UHC with comprehensive, well-grounded responses	Domain-specific expertise with synthesis requirements	Comprehensive synthesis from multiple documents with factual priority
RAG Flow	Factual AI assistant with strict context-only responses	Generic factual accuracy with minimal elaboration	Binary response: accurate answer or "I don't know"

12. Comparative Analysis: Anything LLM vs RAG Flow Performance on UHC/SDG Queries

Technical Configuration Matrix

Table S 7_a : Technical Configuration Matrix

Parameter	*Anything LLM*	*RAG Flow Config A*	*RAG Flow Config B*	*Performance Impact*
Chat Model	Tiny LLama / GPT-4 (API issues)	Llama 3.2 / Gemma 3.2	Llama 3.2 / Gemma 3.2	Model architecture affects UHC data processing
Embedder	Anything LLM Embedder	Nomic-embed-text	Nomic-embed-text	Different embedding strategies impact medical terminology
Reranker	bge-rerank-base	Not specified	Not specified	Re ranking improves healthcare document relevance
Chunking Strategy	1000 tokens	Variable (512/1024)	Variable (512/1024)	Larger chunks preserve UHC context
Overlap	20%	Variable (20%/30%)	Variable (20%/30%)	Higher overlap maintains medical concept continuity
Top K Retrieval	10	10	1	Multiple retrievals vs. focused retrieval for SDG data
Vector Count	21,629	Not specified	Not specified	Large vector space for comprehensive SDG coverage
Similarity Threshold	Not specified	0.2	0.75	Threshold impacts UHC data precision

13. Table S 7_b : Technical Configuration Matrix

Parameter	Config A (Permissive)	Config B (Restrictive)	Rationale
System Prompt (both configs)	You are a factual AI assistant. Your task is to answer questions using only the provided context. 1. Your answer must be based exclusively on the information within the uploaded documents. 2. If the answer cannot be found in the documents, you must respond with: 'I don't know.' 3. Do not add any information that is not explicitly stated in the context.	Same system prompt applied to both Config A and Config B	undefined
Chat Model (Config A)	Llama 3.2	Llama 3.2 (3B) — edge-optimised compact model	undefined
Chat Model (Config B)	Llama 3.2	Same model; parameter tuning only differs	undefined
Embedder	Nomic-embed-text	Platform-native; constitutes acknowledged confound vs. AnythingLLM	undefined
Similarity Threshold	0.2 (A) → 0.75 (B)	Key governance lever: higher threshold = precision over recall	undefined
Vector Similarity Weight	0.8 (A) → 1.0 (B)	Full vector weighting improves semantic matching	undefined
Top N Retrieval	10 (A) → 1 (B)	Single-chunk retrieval in Config B reduces hallucination load	undefined
Temperature	0.8 (A) → 0.0 (B)	Zero temperature enforces deterministic factual output	undefined
Top P	1.0 (A) → 0.1 (B)	Lower nucleus sampling restricts creative generation	undefined
Presence Penalty	0.2 (A) → 0.0 (B)	Removed in Config B to avoid repetition suppression of key terms	undefined
Frequency Penalty	0.3 (A) → 0.0 (B)	Removed in Config B to maintain medical terminology consistency	undefined
Max Tokens	20,000	Consistent across both configurations	undefined

14. Performance Analysis: UHC/SDG Query Results

Table S 7_C : Performance Analysis: UHC/SDG Query Results

Query Type	Anything LLM Performance	RAG Flow Performance	Key Differences
India UHC Index 2022	Expected: 54.3 ; Tiny LLama: Hallucinated Python function (return 0.6)	Llama 3.2: "I don't know" (Config B) Gemma 3.2: Incorrect value (88.918296814)	Anything LLM showed severe hallucination; RAG Flow showed model-dependent accuracy
SDG 3.8 Achievement	Expected: 12 countries (Japan, Norway) Result: Information not found	Llama 3.2: Fabricated country lists (Config A) Gemma 3.2: Vague responses with temporal mismatch	Both platforms failed, but RAG Flow showed configuration sensitivity
OOP Expenditure Trends	Expected: 78% to 52% decrease Result: Knowledge base deficiency identified	Gemma 3.2: Fabricated increasing expenditure data Llama 3.2: Compliant "I don't know" responses	Anything LLM identified root cause; RAG Flow showed mixed compliance
Response Consistency	Multiple identical queries showed 15s vs 500ms variance	Consistent timing within configurations	Anything LLM demonstrated retrieval instability

15. Critical Performance Insights

Table S 7_d: Knowledge Base Adequacy Assessment

Platform	KB Utilization	Data Gap Identification	Response to Missing Data
Anything LLM	Identified KB contains "high-level overview" but lacks "granular, numerical data"	**Superior diagnostic capability** - correctly identified root cause of failures	Appropriately refused to hallucinate when data absent
RAG Flow	Limited analysis of KB adequacy	**Minimal diagnostic feedback** on data availability	Model-dependent: Llama 3.2 compliant, Gemma 3.2 prone to hallucination

16. System Reliability and Consistency:

Table S 7_e : **System Reliability and Consistency:**

Metric	Anything LLM	RAG Flow	Clinical Significance for UHC
Response Consistency	**CRITICAL ISSUE**: Same query yielded different responses (15s vs 500ms)	Consistent within configurations	Inconsistency undermines healthcare decision-making reliability
Hallucination Risk	TinyLLama: Severe (Python function generation)	Configuration-dependent: High in permissive, low in restrictive	Healthcare hallucination poses patient safety risks
Error Recognition	Excellent: Identified "garbage in, garbage out" principle	Limited: Model-dependent awareness	Self-diagnostic capability crucial for medical AI systems

17. Inter-Rater Reliability Details

Table S8: Inter-Rater Reliability Details (Fleiss' κ = 0.82)

Coding Dimension	κ	Agreement	Coder Count	Notes
Fleiss' κ (overall)	**0.82**	Strong agreement	3 independent coders	200 query-response pairs
Factual accuracy (4-point scale)	**0.79**	Substantial	3 coders	Calibration on 20 pilot pairs before main coding
Hallucination presence (binary)	**0.86**	Strong	3 coders	Binary coding: hallucination present/absent
Protocol adherence (binary)	**0.84**	Strong	3 coders	'I don't know' appropriate vs. inappropriate
Response consistency	**ICC per condition**	See Table S4b	Within-condition	20 repetitions per condition (not inter-rater)

Note:

- Three independent coders assessed each of the 200 query-response pairs. Calibration was conducted on 20 pilot responses before main coding commenced.
- Fleiss' κ=0.82 represents 'strong' agreement (Landis & Koch, 1977: κ>0.80=strong).
- Ground-truth values for numerical queries drawn from WHO UHC Service Coverage Index reports and SDG 3.8 monitoring data.
- Coders were blinded to platform condition during scoring.

18. Reproducibility Statement: Hardware, Software, and Experimental Controls

Component	Specification	Rationale / Reference
Processor	AMD EPYC 7502P (32 cores, 2.5 GHz base)	Sufficient parallel processing for simultaneous RAG pipeline evaluation
Memory	128 GB DDR4 ECC RAM	Required for 27B parameter model (Gemma 3) inference without swap
GPU	NVIDIA A100 40 GB	Model inference; A100 selected for FP16 mixed-precision support
Storage	2 TB NVMe SSD	Vector database (21,629 segments) + model weights storage
Network	1 Gbps dedicated	Local-only; no external API calls during evaluation
Operating System	Ubuntu 22.04 LTS	Stable LTS release; confirmed Docker 24.0.5 compatibility
Containerization	Docker 24.0.5 (isolated networks per platform)	Eliminates library version conflicts (Merkel, 2014)
Model Serving	Ollama 0.1.23	Unified inference API for both Llama 3.2 and Gemma 3
Vector Database	Chroma DB 0.4.15 / FAISS 1.7.4	Platform-native; AnythingLLM=Chroma, RAGFlow=FAISS
Embedding AnythingLLM	all-MiniLM-L6-v2 (sentence-transformers)	Platform default; constitutes acknowledged confound
Embedding RAGFlow	Nomic-embed-text	Platform default; different semantic space to MiniLM
Evaluation period	2025 (specific dates available in GitHub repository)	Full experimental logs archived in GitHub
GitHub repository	https://github.com/vinithaFPM23005/-FP05-Bridging-the-Digital-Divide-in-Public-Health.git	Docker installation scripts, system prompts, configuration files

- All experiments were conducted on dedicated local institutional hardware at FORE School of Management, New Delhi (2025).
- No cloud APIs were used; all models ran locally via Ollama to prevent data transmission to external servers.